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Designing for Digital Health: A CeHRes Roadmap Application for Care

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Coordination in the Philippines

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8

CCS Concepts: • Human-centered computing → Accessibility design and
evaluation methods; Usability testing; User models;

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• General and reference → Surveys and overviews; • Hardware → Touch
screens; • Information systems → Enterprise

10

11

information systems; • Security and privacy → Privacy protections; •
Applied computing → Health care information systems. 12

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14

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1 Introduction

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Digital health interventions have emerged has as a notable cornerstone
for the intelligent deployment of modern 23

healthcare infrastructure, offering great transformative potential to
drive policy, elevate care quality, and alleviate

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inflationary operational costs [6, 7]. Nonetheless, the deployment of
sustainable Health Information Systems (HIS) in

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developing nations remains consistently burdened by fragmented
infrastructures, outdated adoption strategies, and 27

critical deficiencies in governance frameworks [5, 23, 30]. Across
Southeast Asia, the primary operational challenges 28

centers around bridging accessibility gaps within highly uncoordinated spaces [21]. Furthermore, in local Philippine

context, macro-level systemic issues take shape as underutilized digital records and overly convoluted data extraction

mechanisms, demonstrating that a significant portion of HIS shortcomings are rooted in a continued lack of attention

to human and contextual factors as opposed to the usual hardware or software constraints [10, 17].

Human-centered pain points are reasonable evidence even within highly advanced institutional settings. As such,

this study investigates the care coordination ecosystem of a private tertiary training hospital located in the Philippines.

Historically, the institute was a leading pioneer in transitioning to a digitized environment; however, there exists a severe disconnect between the hospital's backend administrative capabilities and its public-facing patient care coordination mechanisms. This architectural gap forces out-patients to navigate an ineffective user experience best characterized by

operational vagueness and high logistical friction prior to the patient's clinical consultation proper.

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Such an institutional vulnerability is further worsened by deeply ingrained local patient behavior. Where international 54 healthcare frameworks that mandate primary care providers to serve as the formal 'gatekeepers' and coordinators of the 55 patient journey [24], local Filipino patient populations consistently bypass primary care levels to directly consult with

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their specialists of choice [14]. Labelled "specialist-mindset", the decision-making frequently leads to high occurrences

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of "physician-hopping" leaving patients to personally shoulder the administrative burden of searching for healthcare 59 providers and negotiating schedules [3, 14]. Furthermore, patient-clinician communication channels such as Rakuten 60 Viber or standard SMS text manually managed by individual clinician-secretary pairs [25]. While a cultural norm,

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such ad-hoc methods lead to a distributed human burden that is simply unsustainable, further resulting in frequent

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miscommunications, uncoordinated schedule changes, and heightened risks of appointment mismanagements [3, 25]. 64

To address these gaps, this study details the design, development, and deployment of a multi-interface digital 65 health intervention fundamentally grounded on the CeHRes (Centre for eHealth and Wellbeing Research) Roadmap

66

2.0 [13]. By synthesizing HCD activities with structured business modeling and continuous evaluation cycles, the

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framework ensures a complete structural alignment between the technology, the users, and the organizational context 69

[13, 33]. With the framework as guidance, this study systematically explores three research questions: (1) What specific 70

patient experiences are indicative of an ineffective experience within the medical institute's existing care coordination

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structure? (2) How can the pillars of Human-Centered Design be leveraged to construct a localized, evidence-based care

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coordination platform? (3) How does stakeholder-driven development directly influence the operational success and 74

subjective user experience of a new care coordination system within a resource-constrained clinical environment? 75

By answering these questions, this study provides three notable contributions to the digital health and human

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computer interaction body of literature:

77

78

- Methodological: The study documents a complete traversal of all five interrelated phases of the CeHRes 79

Roadmap 2.0, successfully executing a live deployment of the resulting prototype to address an operational

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implementation gap explicitly cited in prior eHealth literature [9].

82

- Practical: The study delivers a fully functional, department-agnostic care coordination platform comprising 83

integrated interfaces for patients, medical secretaries, clinicians, and system administrators, complete with a 84

physical on-site kiosk interface that establishes a secure digital handoff via dynamic QR code generation.

85

86

- Empirical: The study presents a robust mixed-method evaluation combining a two-week Simple Remote 87

Exploratory Test (SRET) with a summative User Experience Questionnaire (UEQ) deployment ($n = 14$), supplying 88

empirical evidence of a definitive qualitative shift from issue-focused reporting to high user satisfaction across 89

distinct pragmatic and hedonic scales [16, 26].

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The remainder of the study reads as follows:

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2 Related Work

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2.1 Infrastructure and Human Factors in Health Information Systems

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The trajectory of the evolution of HIS is fundamentally represented by the transition from manual record management

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to integrated digital clinical workflows [5]. Prior works establish the successful adoption of modernizations depend on 99

a complex interaction between technological, organizational, environmental, and human-factors, including a baseline 100

computer literacy and interface "user-friendliness" [5]. A robust HIS provides great utility to its stakeholders through

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ease of access, cost reduction, opened lines of communication, and the facilitation of a novel mode of healthcare delivery

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[5].
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CeHRes Roadmap for Care Coordination in the Philippines 3
105
However, in developing countries, the implementation of such sustainable
health informatics continues to face context 106
specific barriers [23]. Empirical reviews of similar implementations
consistently document low governance scores 107
throughout, indicative of deficiencies within specific HIS deployment
strategies [30]. Resource-limited environments
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such as Bangladesh and various Southeast Asian nations are severely
impacted by fragmented and non-digitized
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workflows, lack of training, mismanaged investments, and an absence of
centralized medical databases [21, 23]. Such 111
challenges directly mirror the digital health ecosystem of the
Philippines, with underutilized electronic health records, 112
complex data layers, and rigid infrastructure preventing the deployment
of comprehensive national frameworks [10, 30].
113
This fragmentation becomes highly evident when mapping outpatient care
coordination and triage workflows.
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Within local context, patients regularly bypass primary care providers,
opting instead to navigate their healthcare 116
journey by directly consulting clinical specialists [14]. Such behavior
leads to widespread operational inefficiencies, 117
creating a culture of “physician hopping” due to the low societal
prioritization of family medicine [14]. While national
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regulatory bodies have issued directives attempting to position primary
care levels as the central coordinator and initial
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point of contact within a patient’s healthcare journey [24], the lack of
overall automated, enterprise-grade navigation 121
forces institutes to rely on informal, human-managed communication
channels like Rakuten Viber or standard SMS [25]. 122
Due to the reliance on a distributed human burden for appointment
management, the introduction of severe process
123
bottlenecks, scheduling, and poor patient health outcomes [3, 25].
Addressing these deficiencies would require and
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approach actively centered around human and contextual factors throughout
the design and development lifecycle 126
[10, 17].
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2.2 Human-Centered Design for Specialized Digital Health Interventions

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Human-Centered Design represents a rigorous, innovative problem-solving approach that prioritizes the explicit needs,

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constraints, and mental models of stakeholders throughout the software development process [17, 33]. Historically, the

application of HCD principles within health informatics has demonstrated that incorporating user needs is a critical

prerequisite to improving overall clinical adoption, systematic trust, and population-wide health outcomes [17, 22].

135

Through the active involvement of end-users in interactive prototyping loops, developers would be able to ensure that

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the underlying design logic accurately matches stakeholder needs and local realities [11, 33].

138

The utility brought by user-centered design (UCD) and co-creative methodologies is well-documented across distinct

clinical contexts [11]. Watchler et al. [32] leveraged iterative focus groups to extract user requirements for software

140

targeting depression, ensuring the application resonated positively with patient mental scales. On a similar note,

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Woodard et al. [34] engaged with clinicians and patients to refine and optimize a specialized decision support system for

oncology survivors, proving that iterative feedback loops are a crucial step to making complex medical information

accessibly to non-clinical audiences. Furthermore, prior works highlight the efficacy of ethnographic observations,

145

laboratory simulations, and contextual usability testing towards building functional prototypes for complex conditions

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such as asthma, tuberculosis, and HIV management, even within resource-constrained environments [2, 4, 33].

In the age of documented successes, there exists a critical gap in digital health literature. A vast majority of existing

public-access digital health interventions (DHIs) are condition-specific platforms explicitly built for isolated patient

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populations managing chronic illnesses [11, 22]. There is currently very little attention towards developing multi-role

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institutional systems that seek to unify separate operational needs of patients, medical secretaries, and clinicians

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within a singular care coordination framework. Prior works validate that ignoring the discussed workflows adds immense logistical burdens to clinical workflows [3, 33]. Consequently, this study fills this gap by detailing a centralized,

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department-agnostic system architecture that treats care coordination not as a single-user app, but as a multi-interface ecosystem.

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2.3 Methodological Frameworks and Evaluative Paradigms in eHealth

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To ensure software viability and reduce deployment failures, health informatics research have to be anchored on

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strutured development frameworks [13]. Standard software engineering practices such as Agile or Design Thinking

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165

priotized accelerated software development cycles and quick-fire

ideation. While these are effective for consumer-grade

software, they lack a specialized multi-stakeholder and domain-specific alignment needed for clinical deployment [13].

In contrast, the CeHRes Roadmap offers a holistic, eHealth-specific framework, integrating human-centered activities

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with formal business modeling across five highly interrelated phases:

Contextual Inquiry, Value Specification,

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Design, Operationalization, and Summative Evaluation [13, 31].

171

Through embedding continuous and iterative cycles between each phsae, the framework guides the translation of

medical protocols into intuitive, task-oriented support, whilst actively addressing the contextual reality of scattered

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infrastructures [13, 33]. Prior works have successfully empoloyed the roadmap to map stakeholder mental models and

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ensure organizational alignment [33]. However, many existing studies fail to traverse the framework in full, omitting

the final phase of a live system deployment and its summative evaluation [9]. This study positions itself to address that

methodological gap by detailing a complete execution of the full CeHRes Roadmap culminating in a containerized and

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live deployment.

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Lastly, the iterative and summative evaluation of modern health
technologies requires a shift away from ad-hoc 181
mechanisms into standardized and reliable evaluation instruments [27,
34]. To address this, this study implements 182
the User Experience Questionnaire (UEQ) as a mathematically sound
framework for subjective quality assessment
183
[1]. The UEQ hierarchy evaluates user experience (UX) across six
distinct, semantic scales, splitting the metrics into
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pragmatic quality (comprising Perspicuity, Efficiency, and Dependability)
and hedonic quality (comprising Stimulation 186
and Novelty), anchored on a global Attractiveness baselines [16]. Through
the implementation of this standardized 187
toolkit alongside exploratory testing cycles, the evaluation paradigm
treats UX as a dynamic and evolving process
188
[16, 26]. Providing a transparent, benchmarked assessment of how
stakeholder satisfaction changes as a digital health
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intervention progresses from prototyping to live deployment [12, 26].
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3 Methodology
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3.1 Co-Design Framework: Traversal of the CeHRes Roadmap
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This study adopts a research paradigm focus on user-centered and
iterative software development [18, 28]. Briding 197
the operational gap between institutional information systems and
outpatient care coordination workflows, the 198
development process fully traverses the five highly interrelated phases
of the CeHRes Roadmap 2.0, connected by
199
continuous evaluation cycles [13].
200
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(1) Contextual Inquiry: Grounding the system requirements in stakeholder
context, the existing outpatient workflows
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203
was documented through interviews with domain experts paired with a semi-
structured user-requirements 204
definition survey distributed to the target outpatient population [13].
These activities exposed an overwhelming 205
reliance on informal messaging channels to negotiate appointment
scheduling due to the complete lack of
206

integration with the institute's public-facing interfaces [13].

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(2) Value Specification: Qualitative themes and infrastructural deficits identified during the contextual inquiry were

aggregated into a formalized requirements traceability matrix (RTM) [13].

Specific stakeholder value were

translated into measurable technical requirements, immediately mandating a live clinician directory, automated

212

document generation, and a physical kiosk for handoff capability [13].

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(3) Design: Development advanced from low-fidelity static wireframe prototypes ideated with stakeholders in

Whimsical to a fully functional high-fidelity prototype [13]. Refinement throughout this phase was explicitly

driven by stakeholder feedback loops using a specialized unmoderated testing protocol.

217

(4) Operationalization: Progressing the high-fidelity prototype to production, a stable Minimum Viable Product

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(MVP) was defined, containerized, and deployed within a sandboxed environment on the Railway cloud platform

to completely isolate web traffic from live hospital workflows [13].

221

(5) Summative Evaluation: The final phase saw the execution of an unmoderated two-week Simple Remote Ex

222

ploratory Test (SRET) to monitor real-time issue resolution, followed by the deployment of the UEQ [13, 26].

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3.2 System Architecture and Technical Engineering

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The resulting DHI was engineered as a multi-interface, decoupled web application with four levels of operational

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responsibility: Patients, Secretaries, Clinicians, and System

Administrator. The technical architecture is built around a

containerized development stack setup via Docker to enforce cross-service boundary security and reduce cross-site

scripting vulnerabilities with operational boundaries and constraints mapped in Figure 1.

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The application layers are primarily segregated into three components:

232

233

- Frontend Interfaces: Developed on the React-Vite framework to guarantee high client-side rendering speeds 234 and fluid responsiveness across both desktop and mobile web-browsers.

235

- Backend Engine: Developed on Python leveraging Flask, with the API layer implementing an object-relational

236

237

mapping tier to securely execute transactional database commands.

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- Database: Built on a management PostgreSQL relational instance handling the structured entity records. 239

The database design is anchored by three primary relational entities:

PATIENT (non-diagnostic profiling and

240

authentication credentials); CLINICIAN_SCHEDULE (holding weekly schedule of clinician split into AM/PM time

241

242

pairs); and APPOINTMENT (relational mapping of foreign keys linking patient IDs, time-slots, and real-time status 243 changes).

244

245

In order to integrate physical walk-in users with the digital environment, an on-site touch kiosk was developed. 246

Set to be implemented within a system containing a projected capacitive touch terminal running Windows 10 IoT 247

Enterprise, the kiosk operates on a shared frontend codebase. When a walk-in patient selects a clinician who is currently

248

unavailable, the terminal generates a real-time QR code handoff. Allowing the patient to seamless transfer their session

249

250

and complete the multi-step appointment booking process securely on their mobile device. The transaction pathways 251

and backend operations are documented in Figure 2

252

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3.3 Formalization of the Triage Engine

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Aimed at mitigating the friction caused by local "specialist-first" consulting bias, the digitized triage engine was 256

engineering into the core navigation component [14]. Evolving form a static and arbitrary symptom mapping schema,

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the second-generation engine implements an additive diagnostic heuristic adapted from Friedman's Problem-Oriented

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Medical Diagnosis framework [8]. The functional path executing this is mapped visibly in Figure 3. 260

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Fig. 1. Multi-role system boundaries and interface use cases for the care coordination platform.

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The engine treats the incoming patient as an immutable data tuple ($\langle \text{id},$

$\text{age}, \text{sex} \rangle$, where id is a symptom identifier selected 299

from the bounded patient homunculus diagram, age is an ordinal age-range

modifier, and $\text{sex} \in \{\text{male}, \text{female}, \text{decline}\}$ rep 300

resents biological sex [8]. The heuristic evaluates the candidate

clinical specialty score using the following formulation:

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302

303

$\text{Score}(\text{id} \mid \text{id}, \text{age}, \text{sex}) = \text{score}(\text{id}, \text{age}) + \text{score}(\text{id}, \text{age}, \text{sex}) + \text{score}(\text{id}, \text{age}, \text{sex})$

304
 where s denotes a target medical specialty, $w(s, s)$ represents the
 empirical base likelihood weight mapping a
 305
 symptom to a specialty independent of demographic factors, $w(s, s, s)$
 is the additive age modifier adjustment, and
 306
 307
 $w(s, s, s)$ is the additive biological sex modifier adjustment [8].
 308
 The triage component computes this score across all specialties currently
 mapped within the system [8]. The highest 309
 scoring specialty is passed directly to the clinician directory interface
 as a query parameter to automatically filter the
 310
 visible the clinicians:
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 (a) Level 0 DFD: System boundary and macro data flows. (b) Level 1 DFD:
 Decomposed internal processes and data stores. 330
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 Fig. 2. Data Flow Diagrams outlining enterprise architectural pipelines
 and transactional records boundaries. 332
 333
 334
 335
 $s[Score(s | s, s, s)]$
 $s_{referred} = \arg \max$
 336
 337
 All computed values are processed in applicaiton memory and explicitly
 restricted from being written to permanent 338
 storage or appended visibly to URL strings, ensuring further compliceance
 with human subject data boundaries. 339
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3.4 Mixed-Method Evaluation Protocol

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The evaluation framework utilizes a mixed-method framework aimed at capturing user experience as a dynamic process. 343

This paradigm consists of a two-week qualitative feedback cycle followed immediately by a quantitative metric baseline

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[16, 26].

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The qualitative evaluation saw the Simple Remote Exploratory Test (SRET) across two one-week cycles [18]. Participants 347

were issued the necessary materials and assigned high-level, open-ended operational goals [18]. This approach 348

forced users to rely on natural mental models rather than rigid step-by-step instructions [18]. Real-time UX hiccups

349

and system bugs were logged dynamically via Google Forms and iteratively resolved between cycles [18].

350

351

Immediately after, summative evaluation was administered, utilizing the standardized UEQ toolkit to acquire the 352

multi-dimensional metric breakdown [12, 16]. The questionnaire maps 26

bipolar semantic items across a seven-stage 353

differential scale, loading directly into six overarching dimensions of system design quality [12, 16]:

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355

(1) Attractiveness: Global systemic impression baseline (6 items) [16].

356

(2) Pragmatic Quality Scales: Comprising Perspicuity (ease of learning; 4 items) [16], Efficiency (cognitive speed; 4 357

items) [16], and Dependability (perceived user control; 4 items) [16].

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(3) Hedonic Quality Scales: Comprising Stimulation (engagement; 4 items) [16] and Novelty (perceived design

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360

innovation; 4 items) [16].

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In order to ensure complete consistency during statistical aggregation, the evaluation medium enforces standardized

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Inconsistent and Identical Answers heuristics [26]. Responses are discarded if the item-distance within a single scale 364

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Fig. 3. Activity Diagram tracking the evidence-based routing and appointment booking execution loop.

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flags an outlier threshold of >3 units, or if a participant submits more than 15 identical scalar answers across the

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instrument [26]. The internal scale consistency is continuously monitored through the computation of Cronbach's

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408

alpha (α) coefficient [26].

409
410

$\alpha = \frac{\sum_{i=1}^n \sigma_i^2}{\sum_{i=1}^n \sigma_i^2 + \sigma_e^2}$

411

$1 + (\alpha - 1) \cdot \frac{\sigma_e^2}{\sigma_i^2}$

412

where $\bar{r}$ represents the precise item count allocated per scale and $\bar{r}$ denotes the mean inter-item correlation matrix computed across the dataset [26]. Scale means falling within the interval $[-3.0, +3.0]$ are evaluated, where a mean score $\bar{r} > +0.8$ mathematically establishes a positive evaluation [26].

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4 Results

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The human-centered software engineering paradigm yielded a fully deployed, production-ready care coordination

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ecosystem supplemented by an empirical baseline of its operational, in-context performance. Reflective of the intertwined

420

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nature of the CeHRes roadmap, the results track the system's progression from quantitative psychometrics mapping to

longitudinal qualitative cycle progressions and high-fidelity deployments.

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4.1 Quantitative Psychometrics Dimensions

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Summative verification of the deployed system architecture was established using the standardized UEQ analysis toolkit

[12, 16]. The evaluation tool was administered to the active stakeholder testing group ($n = 14$) immediately after live

428

execution windows to evaluate immediate user impressions across 26 bipolar semantic items [16, 26]. The calculated

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scale metrics, variance limits, and internal consistency parameters are detailed in Table ??.

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432

Table 1. Evaluation Metrics Derived via the Standardized UEQ Data Analysis Tool ($n = 14$) [26].

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UEQ Scale

Mean ($\bar{r}$)

Std. Dev. (σ)

User Agreement

Cronbach's α

Attractiveness

1.74

0.85

Medium

0.86
 Perspicuity
 1.84
 0.75
 High
 0.60
 Efficiency
 1.88
 0.87
 Medium
 0.79
 Dependability
 1.50
 0.95
 Medium
 0.80
 Stimulation
 1.36
 1.00
 Medium
 0.89
 Novelty
 0.55
 1.11
 Low
 0.84

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The system evaluated strongly positively ($\alpha > +0.8$) across five of the six core scales [26]. Pragmatic quality

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dimensions exhibited exceptional ratings, represented by low cognitive friction during objective executive (Efficiency 445

$\alpha = 1.88$) and system-wide learnability for first-time users (Perspicuity $\alpha = 1.84$) [16, 26]. Internal scale consistency met 446

the thresholds required for software verification, with all calculated Cronbach's alpha (α) values safely at or exceeding

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the reliable metric baseline of 0.60 [26].

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Further contextualizing the platform's UX quality, scale means were put against the UEQ benchmark dataset as 450

illustrated in Figure 4.

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(a) Scale means distribution. (b) Benchmark comparison.

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Fig. 4. Quantitative user experience trajectories plotted against the global UEQ reference dataset [12, 26].

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Five of the six scales landed cleanly within the Good performance window, establishing that the human-centered

approach ranks within the top 25% of benchmarked consumer and enterprise software products globally [12]. In contrast,

the lower and more neutral metric observed for the Novelty scale ($\mu = 0.55$) is representative of a highly predictable and

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methodologically reasonable result within the context of the health informatics domain [26]. For systems hosting a

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medical scheduling and triage engine, system predictability, interface control, and strict clarity are quite prioritized

over novel, experimental, or plain unfamiliar layout aesthetics [16].

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4.2 Longitudinal Qualitative Progression

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The qualitative evaluation phases mapped the progressive trajectory of user behavior, capturing direct impacts of

co-creation activities to systematically answer the initial research questions.

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482

- Addressing Ineffective Coordination: Initial survey data ($n = 26$) confirmed prudent process hurdles

within local institutions' traditional care coordination practices.

Overwhelming, outpatients reported that

484

physical concierges were consistently congested, directly causing routing delays. Moreover, over 30 unique

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486

reports document an insecure, manual reliance on informal digital mediums such as Rakuten Viber or SMS for 487

negotiating clinician availability, further leading to frequent appointment appointment mismanagement and 488

distributed burden [3, 25].

489

- Impact of Stakeholder-Driven Development: The two-week Simple Remote Exploratory Test (SRET)

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491

provided empirical proof that iterative co-creative cycles can nurture direct software quality improvements 492

[13, 18]. During SRET Week 1 ($n = 12$), participants logged high success rates for macro goals (11/12 for triage, 493

12/12 for appointment booking), but uncovered 21 unique technical and usability issues [18]. Such roadblocks

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were processed through the action pipeline and patched prior to the next cycle [18].

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496

By the following week ($n = 14$), objective completion rates scaled up to 13/14 across all mobile and document 497

generation fields [18]. Notably, participant feedback shifted from critical error reporting to expressions of high 498

system satisfaction, proving that consistent engagement with stakeholder testing groups positively impacts

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software refinement cycles [18].

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4.3 Technical System Implementations

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In order to address the need for localized and evidence-based care coordination platforms (RQ2), the high-fidelity

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prototype was able to achieve full operational deployment on the Railway production platform. Selections from the

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completed interfaces are organized in Figure 5.

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(a) Patient triage map. (b) Slot picker viewport. (c) Secretary dashboard. (d) Kiosk QR handoff.

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Fig. 5. High-fidelity interface implementations of the deployed multi-role care coordination platform across distinct user-role

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boundaries.

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As detailed in Figure 5, the core components achieve operational functionality across all targeted domains: 522

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(1) Patient-Facing Component: Implemented the interactive homunculus body map (Figure 5a) to drive the multi 524

dimensional additive heuristic. The computed score successfully passes query parameters to filter the clinician 525

directory, routing the patient directly to the integrated slot picker (Figure 5b) displaying real-time clinician

526

availability.

527

528

(2) Clinician-Secretary Component: Consolidated once fragmented communication channels into a central, unified 529

user interface (Figure 5c). Secretaries can update profiles, generate documents, and update appointment statuses 530

in real-time with audit trails.

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(3) Kiosk Application: The kiosk application operates on a shared responsive grid layout. For walk-in patients

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selecting an unavailable clinician, the terminal generates a QR code (Figure 5d), enabling a touchless state 534

handoff to continue the scheduling workflow on the user's mobile device.

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5 Discussion

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The empirical and qualitative findings of this study demonstrate a distinct viability of leveraging HCD frameworks to

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overhaul fragmented clinical care coordination workflows within developing healthcare infrastructures. This section

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will evaluate the study's findings against broader digital health literature, critiquing the co-creative methodology as a 542

software engineering paradigm, and address the limitations of the study.

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5.1 Socio-Technical Implications for Local Digital Health Adoption

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Prior works consistently identify poor strategy, low stakeholder alignment, and a lack of care for human factors as the

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primary failure points for HIS implementations [10, 30]. Through the successful live production deployment scoring

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in the the upper quartile of the global UEQ benchmark dataset, this study presents a valuable proof-of-concept for

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overcoming these implementation barriers [12, 26]. With regional

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evaluations often flagging low governance scores due to a rigid top-down deployment protocols [30], the results confirm a reliable care coordination platform integrating

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natively into an institution's existing workflow ecosystem heavily boosts user compliance, workplace productivity,

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and overall system trust [19, 22].

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Moreover, the system design directly confronts the unique socio-cultural realities of the local population. As opposed

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to ignoring the prevalent "specialist-first" consulting mindset and the corresponding lack of primary-care gatekeepers

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[14], the platform's multi-interface architecture actively accommodates this mindset through the decentralized clinician

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559

directory and on-site QR handoff mechanics. This approach works towards bridging generational technology gaps and

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directly reduces the scattered burdens that build mistrust in healthcare technologies [20]. Consolidating these formerly

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disjoint communication mediums into a singular, traceable transactional framework demonstrating that software safety

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and efficiency can be co-engineering without disrupting ingrained patient consulting behaviors.

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5.2 Reflection on the CeHRes Roadmap

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Critically traversing all five phases of the CeHRes Roadmap 2.0

positioned this study to treat UX as an active, evolving

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software refinement lifecycle as opposed to a static post-development metric [9, 13]. The structured methodology of this

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study directly answers the third research question (RQ3) in regards to the operational influence of stakeholder-driven

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engineering. The qualitative SRET cycle progression clearly presented that actively integrating user feedback loops

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12 Rodriguez and Marasigan

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between development cycles directly drove the positive user sentiment reflected in the quantitative UEQ evaluations

[18, 26].

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The holistic co-design paradigm explicitly underscores the high user ratings observed in the pragmatic dimensions

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of Efficiency ($\bar{\alpha} = 1.88$) and Perspicuity ($\bar{\alpha} = 1.84$), ensuring that the technology accurately aligns with the workflows of

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patients, medical secretaries, and clinical staff alike [16, 26]. Where traditional digital health literature concentrated

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heavily on condition-specific interventions built to serve isolated

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patient populations [15, 34], this study proved that the

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pillars of the CeHRes roadmap are equally effective when applied to

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5.3 Methodological and Technical Limitations

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Despite the presented empirical and qualitative outcomes across the evaluation cycles, several structural limitations

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need to be addressed:

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- Sample Size and Statistical Precision: Although the participant pool ($n = 14$) meets the verified minimum

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baseline for usage of the UEQ, there exist statistical constraints.

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Recruiting a small evaluation group leaves much to be desired for the confidence interval margin, inherently

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- Single-Site and Departmental Boundaries: The nature of implementation narrowed down the deployment

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to a single department, where the system's weighting tables (triageLogic.js) are optimized for a specific outpatient subgroup. The underlying system architecture is department-agonistic, the clinical validity and safety metrics of the triage engine have not been formally measured or verified across diverse medical sub-specialties.

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- Operational Scope Restrictions: To guarantee architectural stability for the primary patient-clinician coordination

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dination flow, a number of advanced system features were scaffolded and deployments deferred; the system

currently contains technical scaffolds for an assisted booking feature for elderly or assisted populations.

- Language and Infrastructure Components: Although the outcomes of initial activities highlighted a relatively

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strong stakeholder preference for multi-language localization layers, the production build was restricted to a

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singular verifiable English layer. A deliberate compromise to eliminate the critical risk of automated or imprecise

translation leading to grave clinical miscommunications during symptom triage. Finally, a fully automated

email confirmation routing had operated at limited capacity due to associated infrastructure costs.

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6 Conclusion

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This study documents the systematic traversal of the CeHRes Roadmap to design, deploy, and evaluate a multi-role

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digital care coordination platform within a private tertiary hospital in the Philippines. Through the establishment of a

human-centered development lifecycle, this work directly addresses three core research questions.

First, in regards to patient experiences within traditional care structures (RQ1), the contextual inquiry verified

that outpatients face severe process hurdles, including an overwhelming operational reliance on informal digital

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communication channels to negotiate clinician availability, creating a highly vulnerable and distributed human burden.

Second, in regards to the implementation (RQ2), the study successfully constructed an integrated, multi-interface

system featuring dedicated components for patients, secretaries, and clinicians, accompanied by a kiosk application

capable of executing state handoff to users' personal mobile devices.
Critically, the platform overrides regional consulting

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biases by employing an evidence-based triage sorting heuristic driven by
an additive multi-dimensional matrix. 624

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Third, in regards to stakeholder (RQ3), the evaluation loops demonstrate
the positive impact of active incorporation 626

of iterative user feedback, driving direct measureable software quality
enhancements. The nature of this co-design 627

paradigm led to high objective completion rates during beta testing, a
definitive qualitative shift from error reporting to

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evident user satisfaction, and exceptionally positive UEQ metrics

positioning the platform's pragmatic usability scales

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in the upper quartile of the global benchmark dataset.

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Cumulatively, this study serves as a valuable proof-of-concept

demonstrating that anchoring DHIs to specialized 632

co-creative roadmaps successfully mitigates the human and organizational
misalignment issues commonly plaguing

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hospital informatics deployments in developing contexts.

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Future researches are recommended to focus on expanding the platform's
current operational footprint through the 636

following avenues:

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- Phased Rollout: Execute a scaled and institution-wide expansion through
engaging representative domain 639

experts from adjacent departments to refine and validate the triage
engine's additive weighting parameters

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across the relevant medical sub-specialties [8, 29].

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- Clinical Accuracy and Dependent Modeling: Conducting clinical
validation studies to measure the exact 643

diagnostic routing accuracy of the problem-oriented logic whilst

deploying a fully functional multi-user dependent 644

booking modes to improve the system's accessibility for elderly or
assisted users [8, 29].

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